

Youngung Jeong

Department of Materials Science and Metallurgy, Changwon National University | yjeong@changwon.ac.kr

+82-55-214-3694 | <https://youngung.github.io> | <https://github.com/youngung>

Education

Hanyang University, BS, Department of Materials Science and Engineering	03/01/2001 – 02/29/2008
POSTECH, MS, Graduate Institute of Ferrous Technology (supervisor: F. Barlat)	03/01/2008 – 02/28/2010
POSTECH, PhD, Graduate Institute of Ferrous Technology (supervisor: F. Barlat)	03/01/2010 – 02/28/2014

Experiences

Assistant, associate, full professor, Changwon National University, ROK	03/01/2017 – Present
Faculty mentor guest, Los Alamos National Laboratory, NM, USA	10/07/2025 – 08/31/2026
Short term visitor, Los Alamos National Laboratory, NM, USA	07/11/2024 – 08/16/2024
Guest Scientist (offsite), Los Alamos National Laboratory, NM, USA	02/01/2022 – 02/1/2024
Postdoc, POSTECH, ROK	12/01/2016 – 02/28/2017
Research Scientist, Clemson university, SC, USA	03/01/2016 – 11/30/2016
Postdoc, NIST, MD, USA	03/01/2014 – 02/29/2016
Research Affiliate, Los Alamos National Laboratory, NM, USA	04/01/2012 – 09/30/2012
Guest Researcher, NIST, MD, USA	06/01/2011 – 12/31/2011

Selected recent publications

• Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations <i>International Journal of Plasticity</i> 191 José Victoria-Hernández*, <u>Y. Jeong*</u> , Dietmar Letzig	2025
• Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel <i>International Journal of Plasticity</i> 181 B. Jeon, S.-Y. Lee, J. Lee, <u>Y. Jeong*</u>	2024
• A crystal plasticity finite element analysis on the effect of prestrain on springback <i>International Journal of Mechanical Sciences</i> , 237 M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, <u>Y. Jeong*</u> ,	2023
• Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading <i>International Journal of Plasticity</i> 147 <u>Y. Jeong*</u> , B. Jeon, C. N. Tomé	2021

Government-funded projects

Mid-career Researcher Program (PI), National Research Foundation of Korea	2023 - 2027
BK21, National Research Foundation of Korea	2020 - 2027
Engineering Research Center, National Research Foundation of Korea	2018 - 2024
General Research Program (PI), National Research Foundation of Korea	2020 - 2023
Virtual engineering platform project, Ministry of Trade, Industry and Energy	2018 - 2020
General Research Program (PI), National Research Foundation of Korea	2017 - 2020

Skills

Constitutive modeling: Phenomenological plasticity and crystal plasticity modeling

Languages: Korean, English

Computer skills: Linux, Git

Programming languages: Fortran, Python (NumPy, SciPy, and matplotlib), Matlab, shell scripts

Experimental Mechanics: uniaxial tension, shear, hydraulic bulge test, biaxial tests, digital image correlation (DIC)

Diffraction experiments: X-ray diffraction, Electron back-scattered diffraction (EBSD), Residual stress analysis

GitHub repositories

Elasto-visco-plastic self-consistent model (private repo) github.com/youngung/evpsc

- Extended visco-plastic self-consistent model to account for elasticity (Δ EVpsc and σ EVpsc)
- Stand-alone calculation capabilities
- Various example shell and python scripts as well as Jupyter notebook files
- User material subroutine (UMAT) of Abaqus/standard solver
- Written primarily in Fortran with Python and shell scripts

VPSC8 and EVpsc (private repo)

github.com/Jeonbohye/VPSC8-dEVpsc

- Combines VPSC8 and EVpsc models
- Numerical implementation of VPSC, Δ EVpsc, and σ EVpsc constitutive model
- Three separate build commands for executables of VPSC8, Δ EVpsc, and σ EVpsc.

In-house Python scripts for texture analysis

github.com/youngung/texture3

- Can plot contoured pole figures and inverse pole figures from discrete orientations
- Written entirely in Python with open-sourced libraries including matplotlib, NumPy and SciPy
- Can generate RGB maps of EBSD (electron back-scattered diffraction) scans

Misc.

- Board member of Korean Society for Technology of Plasticity 2023-present
- Editorial board member of Korean Journal of Metals and Materials 2019-present

Full list of publications

1. Application of Δ EVPSC to interpret mechanical behavior of AZ31B sheet in warm temperature in preparation
Kyoungwon Hwang, Bohye Jeon, Jaeseong Lee, Hyuk-Jong Bong, Jinwoo Lee, Y. Jeong*
2. Elasto-visco-plastic model for deformation of dual phase steels submitted
Anirban Patra*, Y. Jeong*, Carlos N. Tome
3. A cluster-based mean-field crystal plasticity modeling framework to simulate dynamic recrystallization submitted
Y. Jeong*, Carlos N. Tome, Andrea Rovinelli, Laurent Capolungo
4. Calibrating self-consistent crystal plasticity models beyond stress-strain fitting: mechanism-activity features to discover solution families submitted
Jinyeong Yu, Jaeseong Lee, Y. Jeong*, T. Lee*
5. Reverse engineering material behavior using Bayesian inference and finite element analysis on ring-pull test submitted
B. Jeon, C. N. Tome, P. M. Beck, B. Eftinik, A. Talapatra, Y. Jeong*, L. Capolungo*
6. (1012) Twinning, Slip, and Texture in AZ31 Mg under Uniaxial Compression: A Comparative Study of Schmid Factor and Crystal Plasticity Modeling accepted
Metals and Materials International
Jongbin Go*, Y. Jeong*, Myeong-heom Park*, Sangwon Lee, Si Gao, Nobuhiro Tsuji
7. Leveraging Machine Learning and Crystal Plasticity for Efficient Calibration of Yld2004-18p Anisotropic Yield Function 2026
International Journal of Mechanical Sciences, 328, 112039
A. S. Ebrahim, J. Kim, Y. Jeong, T. Park, H. Lim, B. L. Kinsey, J. Ha*
8. Experimental and numerical investigations on the microstructure and tensile deformation behavior in flash-butt welded joints of boron steel and the effects of a post heating condition 2026
Materials Characterization, 238, 116631
Joonoh Moon*, Gyuyeol Bae*, Y. Jeong, Min-Ji Kwon, Jaeseong Lee
9. A Scientific Benchmark for Elasto-Plastic Constitutive Modeling - Part II: Blind Predictions, Calibration Strategies, and Benchmark Results 2026
International Journal of Material Forming, 19:17
Lorenz Maier, Erik Walz, Michael Hofmann, Joana Rebelo Kornmeier, Rongfei Juan, Lian Junhe, Youngeung Jeong, Bohye Jeon, Martin Milch, Alper Guner, Emad Maawad, Wolfram Volk, Christoph Hartmann
10. Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations 2025
International Journal of Plasticity, 191, 104380
José Victoria-Hernández*, Y. Jeong*, Dietmar Letzig
11. Modeling deformation, recovery, and recrystallization of tantalum using a higher order elasto-viscoplastic self-consistent model 2025
Journal of the Mechanics and Physics of Solids, 194, 105925

- I. A. Riyad, B. Clausen, D. J. Savage, Y. Jeong, D. W. Brown, M. Knezevic*
12. A critical discussion of elasto-visco-plastic self-consistent (EVPSC) models 2024
Journal of Materials Research Technology, 33, 7596-7609
 B. Jeon, Y. Jeong*, C. N. Tomé
 13. Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel 2024
International Journal of Plasticity, 181, 104098
 Bohye Jeon, Shin-Yeong Lee, Jinwoo Lee, Youngung Jeong*
 14. Thermal ratcheting of uranium simulated with a thermo-elasto-visco-plastic self-consistent polycrystal model 2024
Journal of Nuclear Materials, 597, 155159
Youngung Jeong*, Carlos N. Tomé
 15. Crystal plasticity finite element simulations on extruded Mg-10Gd rod with texture gradient 2024
Journal of Magnesium and Alloys, 12 , 3409-3430
 Jaeseong Lee, Dirk Steglich, Youngung Jeong*
 16. A comprehensive analysis of cermet design and thermal cyclic stability via elasto-viscoplastic crystal plasticity modeling 2024
International Journal of Plasticity, 179, 104032
 Glenn R Peterson, Youngung Jeong, Carlos N Tomé, Michael D Sangid*
 17. Temperature-dependent behavior of CP-Ti interpreted via self-consistent crystal plasticity simulation 2024
Materials Science and Engineering: A, 890, 145904
 Bohye Jeon, Min-Su Lee, Tea-Sung Jun, Youngung Jeong*
 18. Finite element analysis using elasto-visco-plastic self-consistent polycrystal model for E-form Mg sheet subjected to bending 2023
Journal of Magnesium and Alloys, 11, 1393-1407
 B. Jeon, M. S. Kim, S. H. Choi, Y. Jeong*
 19. A crystal plasticity finite element analysis on the effect of prestrain on springback 2023
International Journal of Mechanical Sciences, 237, 107796
 M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, Y. Jeong*
 20. Reconstructing orientation data from the images of IPF maps and ODF sections extracted from the literature: A data-collection method for machine learning 2023
International Journal of Plasticity, 159, 103467
 L. Kaushik, K.-S. Park, J.-G. Kim, J. Lee, Y. Jeong, S.-H. Choi*
 21. Prediction and validation of stress triaxiality assisted by elasto-visco-plastic polycrystal model 2022
Korean Journal of Metals and Materials, 60, 607-618
 Jinhwa Park, Youngung Jeong*

22. In-situ neutron diffraction study of lattice deformation behaviour of commercially pure titanium at cryogenic temperature 2022
Scientific Reports, 12, 3719
M.-S. Lee, T. Kawasaki, T. Yamashita, S. Harjo, Y.-T. Hyun, Y. Jeong, T.-S. Jun
23. Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading 2021
International Journal of Plasticity, 147, 103110
Y. Jeong*, B. Jeon, C. N. Tomé
24. An efficient elasto-visco-plastic self-consistent formulation: Application to steel subjected to loading path changes 2020
International Journal of Plasticity, 135, 102812
Y. Jeong*, C. N. Tomé
25. Modelling-assisted description of anisotropic edge failure in magnesium sheet alloy under mixed-mode loading 2020
International Journal of Mechanical Sciences, 181, 105680
Y. Jeong*, Dirk Steglich
26. Extension of the VPSC model to account for elasto-visco-plastic behavior using a perturbed viscoplastic approach 2019
Modelling and Simulation in Materials Science and Engineering, 27, 085013
Y. Jeong*, C. N. Tomé
27. Superior tensile fracture strength of hot isostatically pressed TiC–steel metallic composite fabricated by a novel infiltration 2019
Materials Science and Engineering: A, 764, 138260
S. J. Park, Y. Jeong, C. W. Kim, J. H. Lee, S. C. Cho, S. B. Lee, S. K. Lee, D. H. Kim, H. U. Hong*
28. Enhancement in viscoplastic self-consistent FLD prediction model and its application for austenitic and ferritic stainless steels 2019
Metals and Materials International, 25, 1548–1563
Y. Jeong*, T. Manninen
29. A crystal plasticity model for describing the anisotropic hardening behavior of steel sheets during strain-path changes 2018
International Journal of Plasticity,
H. Kim, F. Barlat, Y. Lee, S. Zaman, C. S. Lee, Y. Jeong*
30. A comparative study between micro- and macro-mechanical constitutive models developed for complex loading scenarios 2017
International Journal of Plasticity, 93, 212-228
Y. Jeong*, F. Barlat, C. N. Tomé, W. Wen
31. Uncertainty in flow stress measurements using X-ray diffraction for sheet metals subjected to large plastic deformations 2016
Journal of Applied Crystallography, 49, 1991-2004
Y. Jeong*, T. Gnäupel-Herold, M. Iadicola, A. Kreuziger

32. Texture-based forming limit prediction for Mg sheet alloys ZE10 and AZ31 2016
International Journal of Mechanical Sciences, 117, 102-114
D. Steglich, Y. Jeong*
33. Forming limit prediction using a self-consistent crystal plasticity framework: a case study for BCC fiber textures 2016
Modelling and Simulation in Materials Science and Engineering, 24, 055002
Y. Jeong*, M.-S. Pham, M. Iaidocola, A. Creuziger, T. Foecke
34. Multiaxial constitutive behavior of an interstitial-free steel: measurements through X-ray and digital image correlation 2016
Acta Materialia, 112, 84-93
Y. Jeong*, M. Iaidocola, T. Gnäupel-Herold, A. Creuziger
35. Effect of martensitic phase transformation on the behavior of 304 austenitic stainless steel under tension 2016
Materials Science and Engineering: A, 649, 174-183
H. Wang*, Y. Jeong*, B. Calusen, Y. LiU, R. J. McCabe, F. Barlat, C. N. Tomé
36. Evaluation of biaxial flow stress based on Elasto-Viscoplastic Self-Consistent analysis of X-ray Diffraction Measurements 2015
International Journal of Plasticity, 66, 103-118
Y. Jeong, T. Gnaupel-Herold, F. Barlat, M. Iadicola, A. Creuziger, M.-G. Lee*
37. Application of crystal plasticity to an austenitic stainless steel 2012
Modelling and Simulation in Materials Science and Engineering, 20, 024009
Y. Jeong*, F. Barlat, M.-G. Lee
38. Biaxial Deformation Behavior of AZ31 Magnesium Alloy: Crystal-Plasticity-Based Prediction and Experimental Validation 2012
International Journal of Solids and Structure, 49, 3551-3561
D. Steglich*, Y. Jeong, M. O. Andar, T. Kuwabara